



CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Hanoi, Nov 2025 –

Table of Contents

Contents

I. Record of Changes	3
Definitions and Acronyms.....	3
II. Testing Documentation.....	5
1. Scope of Testing	5
1.1 Target of Test (features & functions).....	5
1.2 Non-functional requirements	5
2. Test Strategy	6
2.1 Testing Types.....	6
2.2 Test Levels	6
2.3 Supporting Tools	6
3. Test Plan.....	7
3.1 Test Environment	7
3.2 Test Milestones	7
4. Test Cases.....	7
4.1 Unit Test.....	7
4.2 Integration Test.....	8
4.3 System Test	8
5. Test Reports	8
5.1 Unit test report	8
5.2 Integration test report	9
5.3 System test report.....	10

I. Record of Changes

Date	A* M, D	In charge	Change Description
5/12/2025	A	QuatBt	Create file
12/12/2025	A,M	TuanCm	Add contents, reports
27/12/2025	M	Team G44	Update test documentation

*A - Added M - Modified D - Deleted

Definitions and Acronyms

Acronym	Meaning
A	Added
M	Modified
D	Deleted
OLP	Online Learning Platform
RESTful	Representational State Transfer
CORS	Cross-Origin Resource Sharing
API	Application Programming Interfaces

CSS	Cascading Style Sheets
JWT	JSON Web Token
OTP	One-Time Password
UT	Unit Test
IT	Integration Test
ST	System Test
JSON	JavaScript Object Notation
CV	Curriculum Vitae
Inc	Incorporated

II. Testing Documentation

1. Scope of Testing

1.1 Target of Test (features & functions)

The tests will cover the OLP-SkillUp features below (primary targets):

- **Course Management:** Course creation, editing, publishing (draft/published states), curriculum organization with sections and lessons, course categories and subcategories, course enrollment and progress tracking.
- **Assessment & Quizzes:** Quiz creation with multiple question types, timer and passing percentage configuration, quiz attempts and scoring, student progress tracking.
- **Student Progress & Enrollment:** Course enrollment, lesson completion tracking, progress percentage calculation, learning history and recently accessed courses.
- **Rating & Reviews:** Course ratings (star-based), written reviews and feedback, rating aggregation and average calculation, student-only rating permissions.
- **Lecturer Application System:** CV submission for becoming a lecturer, application status tracking (Pending/Approved/Rejected), admin review and approval workflow, lecturer profile with bio and expertise.
- **Payments:** Shopping cart functionality, course pricing (free/paid), voucher/coupon system with discount codes, transaction processing, payment method integration, order history.
- **User Management:** Registration, login, profile management, role-based access (Student, Lecturer, Content Moderator), password reset with OTP, token-based authentication (JWT + Refresh Token).

1.2 Non-functional requirements

The following non-functional areas will be validated at a basic/operational level:

- **Compatibility:** RESTful API architecture for cross-platform support, CORS configuration for frontend-backend communication, support for multiple browsers (modern browsers via Vite/React), mobile-responsive design with Tailwind CSS.
- **Security (basic):** JWT-based authentication with token expiration, role-based authorization (Student, Lecturer, Content Moderator, Admin), refresh token mechanism for session management, OTP verification for password reset, input validation and sanitization, secure file upload restrictions.
- **Reliability:** correct error handling and stable recovery for common failure cases.
- **Documentation:** correctness and completeness of user guides and API docs.

2. Test Strategy

2.1 Testing Types

2.2 Test Levels

Type of Tests	Test Level		
	Unit	Integration	System
Function Testing	X		X
Integration Testing		X	X
Performance Testing		X	X
API Testing	X	X	

2.3 Supporting Tools

Purpose	Tool	Vendor/In-house	Version
Test Result	Google sheets(excel)	Google	Latest
Front-end test	Chrome Developer/ Microsoft Edge	Google/Microsoft	Latest
Manage Bugs	Google sheets(excel)	Google	Latest

3. Test Plan

3.1 Test Environment

Purpose	Tool	Provider	Version
Test API	Postman	Postman.Inc	Lastest
Test UT	NUnit	.NET Foundation	3.14.0
Test IT	Chrome Developer/ Microsoft Edge	Google/Microsoft	Lastest
Test ST	Chrome Developer/ Microsoft Edge	Google/Microsoft	Latest

3.2 Test Milestones

Milestone Task	Start Date	End Date
Test Planning	21/9/2025	12/12/2025
Unit Testing Execution	3/11/2025	12/12/2025
Integration Testing Execution	17/11/2025	12/12/2025
System Testing Execution	7/12/2025	12/12/2025
API Testing Execution	2/10/2025	12/12/2025
Test Report Preparation	11/12/2025	12/12/2025
Bug Fixing and Retesting	6/11/2025	12/12/2025

4. Test Cases

4.1 Unit Test

The unit testing phase focuses on verifying the core methods mentioned above. Each unit test follows a structured approach:

- Prepare the test design: Define the required setup, such as object initialization, identifying valid input ranges, and applying techniques like boundary value analysis.
- Implement the test in the IDE: Invoke the target methods according to the prepared design. Each test is executed independently, using mock objects to replace external dependencies. Instead of covering every single line, the tests prioritize the most critical functionalities of the unit under evaluation.
- Validate and record results: Use predefined assertions in the code to confirm whether the output matches expectations. The testing framework executes these assertions, flags any failures, and the outcomes are documented in the corresponding test script.

For more full detail test cases: [Report5.1_Unit Test.xlsx](#)

4.2 Integration Test

Integration testing is performed by combining individual modules and examining them as a complete group. This phase aims to ensure that the interconnected components interact properly and that the system as a whole operates according to expectations.

For more full detail test cases: [Report5.2 Integration Test.xlsx](#)

4.3 System Test

- This testing approach focuses on observing and assessing how a complete system operates. It integrates various management processes to achieve the outcomes users need.
- Test cases serve as blueprints for system testing. They specify the actions performed within the system, the data provided as input, and the anticipated results at each stage of the test.

For more full detail test cases: [Report5.3 System Test.xlsx](#)

5. Test Reports

5.1 Unit test report

Unit testing is conducted to verify the correctness of individual functions and components of the system. Each unit is tested independently to ensure that its internal logic, input validation, and output results function as expected. Any defects detected during this phase are recorded and corrected to ensure the stability of each unit.

The unit testing phase achieved complete success with 312 test cases executed, demonstrating 100% test coverage and 100% successful coverage.

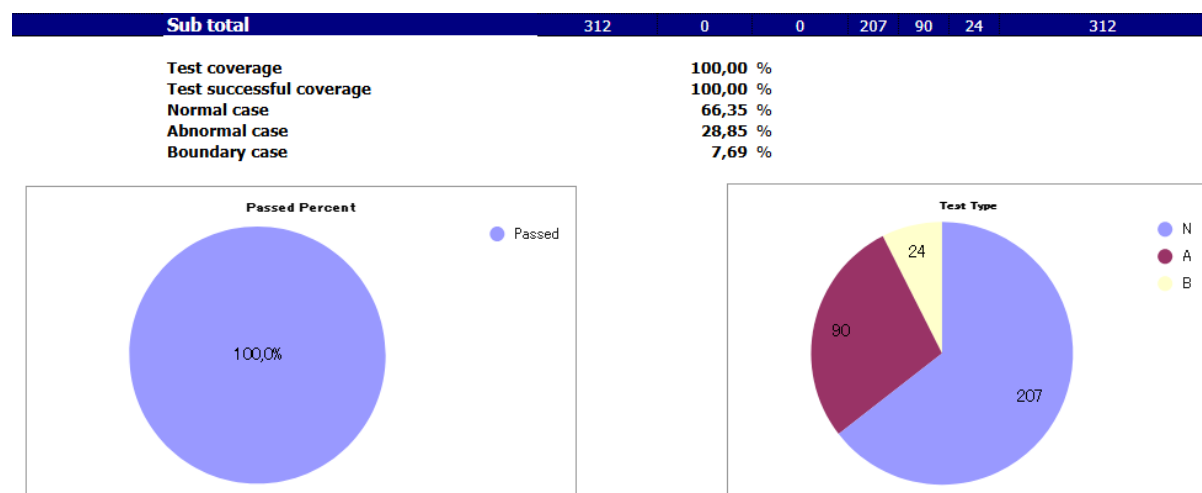


Figure 5.1: Unit test report

5.2 Integration test report

Integration testing will involve incorporating the API into the web app's interface to assess the data representation. The tester will select test cases from the Test Report wherein actions are performed between functions or data is exchanged between modules. If any errors occur during testing, specifically concerning unexpected data, testers will log defects or bugs on the test report sheet and track the failed test case until it is addressed and resolved by developers.

Integration testing was conducted with 104 test cases, All integration points were tested thoroughly with zero failures.

TEST STATISTICS

Project Name	Online Learning Platform – SkillUp	Creator		TuanCM
Project Code	OLP-SkillUp	Reviewer/Approver		
Document Code	OLP-SkillUp_System_Test_Report	Issue Date		3/11/2025
Notes				

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Register	3	0	0	0	3
2	Login by account	3	0	0	0	3
3	Login by google	1	0	0	0	1
4	Forgot password	2	0	0	0	2
5	Logout	3	0	0	0	3
6	View profile	1	0	0	0	1
7	Update profile	2	0	0	0	2
8	Change password	4	0	0	0	4
9	View purchase history	2	0	0	0	2
10	View cart	2	0	0	0	2
11	Add to cart	3	0	0	0	3
12	Remove from cart	2	0	0	0	2
13	Add course	2	0	0	0	2
14	Hide course	1	0	0	0	1
15	Update course - general	9	0	0	0	9
16	View section	2	0	0	0	2
17	Add section	4	0	0	0	4
18	Edit section	3	0	0	0	3
19	Delete section	1	0	0	0	1
20	Create lesson	6	0	0	0	6
21	Create question	4	0	0	0	4
22	Add from bank	2	0	0	0	2
23	View question	2	0	0	0	2
24	Edit question	3	0	0	0	3
25	Delete question	2	0	0	0	2
26	Enroll course	3	0	0	0	3
27	Learning course	3	0	0	0	3
28	Progress tracking	1	0	0	0	1
29	Checkout	5	0	0	0	5
30	Apply voucher	5	0	0	0	5
31	Take quiz	6	0	0	0	6
32	View result	6	0	0	0	6
33	Rating	2	0	0	0	2
34	Email notification	4	0	0	0	4
Sub total		104	0	0	0	104

Test coverage 100,00 %
Test successful coverage 100,00 %

Figure 5.2: Integration test report

5.3 System test report

This testing stage occurs after the functions outlined in the requirements have been completely implemented. We will conduct tests using Chrome and Edge. The primary objective of this test phase is to evaluate the performance of the application when operated by the user on the web environment.

System testing encompassed 174 test cases across seven modules achieving 100% test coverage and 100% successful coverage. Notably, one test case in the Authentication module failed during the first testing round but was successfully fixed and verified in rounds 2 and 3, resulting in all modules passing with zero failures.

TEST STATISTICS

Project Name	Online Learning Platform – SkillUp	Creator		TuanCM
Project Code	OLP-SkillUp	Reviewer/Approver		
Document Code	OLP-SkillUp_System_Test_Report	Issue Date		7/12/2025
Notes				

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Authentication	15	0	0	0	15
2	Workflow of System Moderator	15	0	0	0	15
3	Workflow of Content Moderator	32	0	0	0	32
4	Workflow of Lecturer	63	0	0	0	63
5	Workflow of Student	33	0	0	0	33
6	Workflow of Admin	16	0	0	0	16
7	AI assistant	5	0	0	0	5
Sub total		174	0	0	0	174

Test coverage 100,00 %
Test successful coverage 100,00 %

Figure 5.3: System test report